WHITEPAPER

Moving from SDLC to AI-Driven Software Development Lifecycle (ADLC) to Generate Value

We are living in an era where AI is redefining how we live, and software development is no exception. In fact, AI is transforming the entire Software Development Lifecycle (SDLC). OEMs, analysts and other industry commentators emphasize the products available in the market and the extraordinary benefits they offer. However, the adoption levels are still far below expectations. Many organizations struggle to realize value by adopting AI in SDLC.

Does this mean these tools lack value? Absolutely not. However, it is important to understand that adopting a tool is only one part of the broader transition to AI/Gen AI adoption in SDLC. Organizations must address several other factors within their ecosystem to scale adoption and reap real, ground-level benefits.

This white paper aims to highlight the challenges holding organizations back from realizing the benefits of AI adoption in SDLC and propose methods to address them effectively.

Contents

1. The Evolution of SDLC

Over the decades, SDLC has undergone significant transformation, evolving from a rigid, linear model to a more flexible, iterative one, incorporating technological advancements and changes in market needs. The figure below illustrates the characteristics and challenges associated with each methodology and highlights how later models address the shortcomings of their predecessors.

	Waterfall	Iterative	Agile	Scaled Agile	Desired State (ADLC) Scaled Agile with AI infusion
Features	• Predictability and Simplicity • Works with well-defined requirements	• Evolving documentation. • Continuous testing for early issue detection.	• Just enough documentation. • Continuous integration, testing, and deployment.	• Consistency in documentation. • Systematic and integrated approach with a continuous delivery pipeline.	• AI-Assisted automation of all SDLC activities which includes, but not limited to, automated generation of documents, design, code, test cases, test data, automation test scripts, deployment scripts, etc.
Challenges	• Static documentation created in large monolithic blocks. • Extensive documentation at project start is error-prone. • Late discovery of defects.	• Document synchronization. • Keeping test cases and results consistent and updated across iterations.	• Balancing detail with agility. • Creating and maintaining a robust automation test suite is challenging. • Needs experienced team.	• Maintaining consistent artifacts across multiple teams is a challenge. • Coordinating testing efforts across multiple teams and aligning them is cumbersome.	
Impact	• Higher time to market. • Increased cost and resource allocation. • Customer dissatisfaction.	• Higher operational overhead. • Inconsistent quality of deliverables. • Unpredictable delivery timelines.	• Dependence on human skills and expertise. • Increased initial investment. • Efficiency and quality trade-offs.	• Impact on time-to-market. • Higher dependence on human interventions. • Reduced cross-team efficiency.	• Faster time-to-market. • Reduced dependence on human expertise. • Consistent quality of deliverables.

Despite all the evolutions in SDLC, AI infusion continues to be low across different life cycle stages. Some of the key challenges that organizations face include:

Process Pitfalls: Manual requirements detailing and lack of design standards lead to product gaps and inconsistencies.

Steeper Learning Curve: Poor system documentation slows onboarding.

Complexity Overload: Over-engineering and technical debt hinders performance.

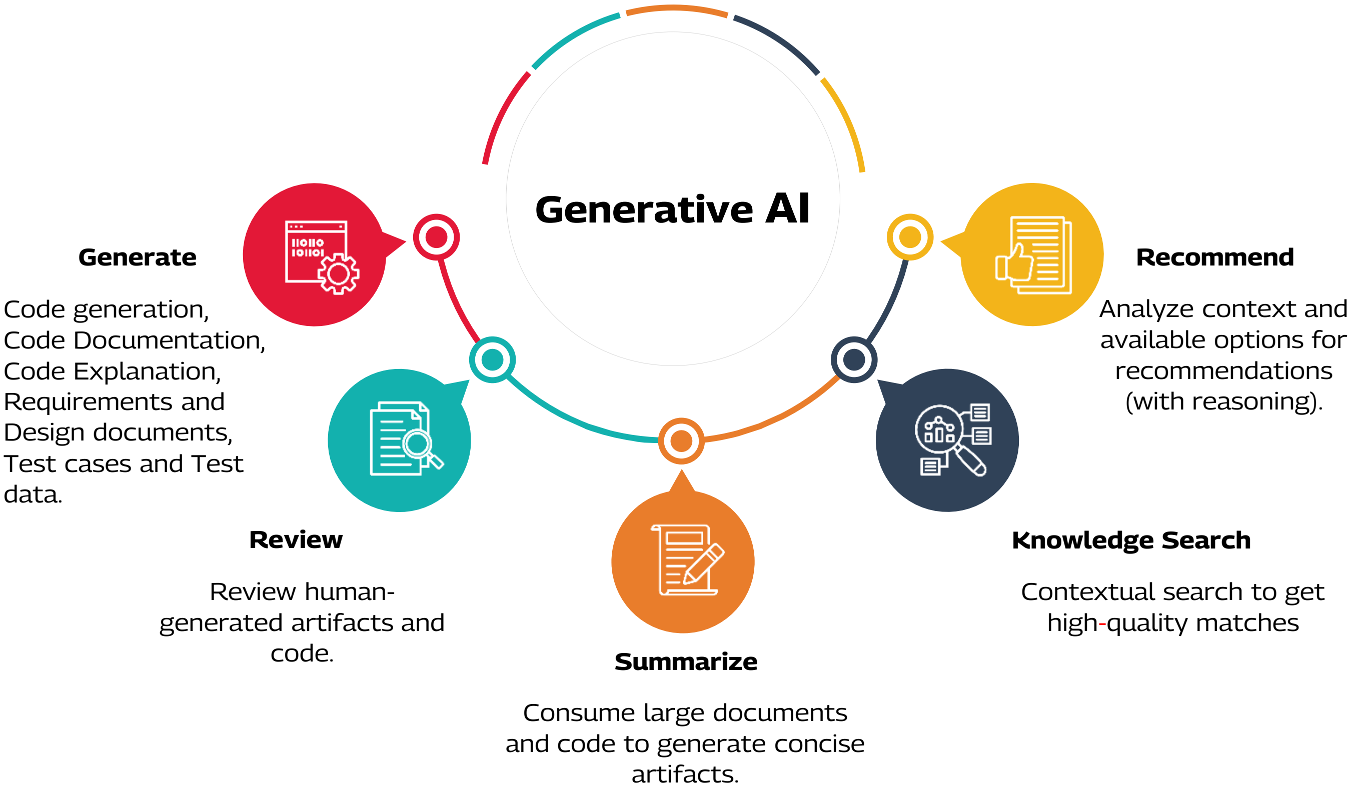
Hidden Costs of Manual Reviews: Manual reviews of artifacts result in higher costs in later stages.

Manually Intensive: Manually intensive tasks and SME dependency pose higher risk.

Time to Market and Cost: These issues lead to increased time-to-market and higher Total Cost of Ownership (TCO).

2. Opportunities Presented by AI/Gen AI in SDLC

The inefficiencies explained in the previous section can be mitigated by various capabilities of AI/Generative AI as depicted below



These capabilities can be applied across various stages of the SDLC, from development to steady-state maintenance, to improve project outcomes. Some of these are depicted below. *(The percentages indicate potential productivity improvements)*

Requirement Analysis	Design	Build	Test
S Generate overview and glossary from BRDs.	R Generate design options from BRDs.	G Write code faster with natural language instructions.	G Generate test cases – both document and code.
Rv Review and check completeness of requirement documents.	K Find similar designs for better reuse.	G Improve code readability with AI-generated comments.	Rv Review test coverage with respect to requirements.
K Find relevant information on a feature across documents.	G Generate architecture design, class diagrams, sequence diagrams etc.	Rv Perform faster and accurate code reviews.	G Generate synthetic data for testing.
Up to **20%**	Up to **15%**	Up to **30%**	Up to **30%**

G Generate Rv Review S Summarize K Knowledge Search R Recommend

3. Preparing Organizations for AI adoption and Benefits Realization

Despite AI/Gen AI presenting several opportunities to address SDLC challenges, improve time-to-market, reduce cost and improve quality, many organizations struggle to realize its practical value. This section discusses the key reasons why this happens and outlines ways of addressing the challenges.

Planning & Execution	Technology Considerations	Commercial Considerations	Stakeholder Management

Challenges

Planning and Execution: Project planning and execution plays a significant role in shaping value realization from AI interventions. Some of the challenges in this area that can plague organizations are:

1. A narrow focus on tool adoption without holistic planning will not result in the anticipated value. Tool adoption is just one of the steps in the process, and not the only step. For example, if a team adopts an AI Pair Programming tool, it will certainly help accelerate the coding phase. However, it will not result in a reduction in the overall time-to-market or TCO because the overall cycle remains unchanged.
2. The lack of a good training environment limits employees' ability to appreciate and adopt such technologies for their benefit.
3. Poorly designed information security architecture and protocols result in higher risks and lower adoption.
4. Implementation partners with limited skills who are unable to demonstrate a strong adoption roadmap through proof-of-value.

Remedies

Planning: The benefits derived by infusing AI into one of the SDLC phases will not materialize unless the entire cycle is replanned. For example, imagine an Agile team with a 4-week iteration cycle where an AI Pair Programmer is implemented to help developers with their coding and unit testing activities. This would probably result in increased developer productivity. However, this improvement alone will not lead to any benefits for the organization. To derive the benefits of faster coding, the subsequent testing and deployment activities also need to be advanced. This will lead to shorter release cycles with the same available bandwidth, which will result in cost savings.

Training: Successful AI adoption hinges on relevant training sessions across the board. Leaders and teams involved in this transformation must be empowered with the necessary knowledge and skills for faster adoption. Hence, a robust training framework is needed to keep the relevant people abreast of the emerging capabilities and techniques.

Partner Selection: When an organization or parts of an organization undergo transformation, it is important to have the right partner. A partner with the experience and capability to handle these tasks, who contributes valuable insights and demonstrates determination in this endeavor, and who practices what they preach rather than just advising customers is critical.

Challenges

Technical decisions and the prevailing landscape significantly influence the ROI from AI interventions in the SDLC:

1. Insufficient investment in technical infrastructure, including AI platforms, necessary AI tools, or a reuse repository, plays a part in impacting value delivery.

2. Uncontrolled usage of LLMs-on-cloud can result in high subscription costs, negatively impacting the overall ROI.

3. Inadequate technical depth among leaders and mid-managers limits their ability to effectively evaluate, select, and adopt the right AI capabilities and tools.

Remedies

1. **Infrastructure Support:** The technology infrastructure must be supportive enough for people to leverage the tools and platforms required for AI infusion throughout the SDLC. Even small incremental changes can deliver good results. For example, an organization already using Microsoft Azure can enable the GitHub Copilot service on their existing instance to take advantage of the tool with a small incremental cost.

2. **Cost Control:** The increasing subscription costs of LLMs and other AI services could potentially erode the ROI from productivity benefits. Hence it is necessary to devise mechanisms to restrict unnecessary usage. In addition, organizations can look for on-premise alternatives that are provided by open-source counterparts

3. **Democratization:** Organizations can democratize AI infusion by providing SDLC copilots to developers and teams, like design copilot, coding copilot, testing copilot, etc. These copilots can be built as nimble solutions encapsulating AI and Gen AI capabilities. Also, these copilots can be seamlessly integrated into existing processes and systems to improve usability and user experience, and to ensure higher adoption levels.

4. **Checks and Balances:** When using LLMs and other AI services from a public cloud, a robust information security policy, checks and balances, checklists, training, and accountability needs to be in place.

5. **User Agreements:** If certain AI models and AI services must be inevitably consumed only from a public cloud, organizations can stitch an indemnity agreement with the AI product/platform vendors to protect themselves from any potential legal issues.

6. **Reusability:** Reusability plays a significant role in the speed of adoption and execution. Therefore, an organization-wide repository for AI reuse, where people can consume and contribute assets, is needed.

Challenges

Another important factor that can minimize or even neutralize the return on investment is poor commercial decision-making. There is a tendency in many organizations to acquire a plethora of tools without knowing what to do with them, or how to use them effectively. This problem only aggravates in federated organizational structures where every team decides what they want. This significantly increases the cost to the company, overshadowing the value derived from these tools.

Remedies

Technology procurement of an organization should be centralized keeping in mind the needs of various departments, teams, budgets, and costs. Many organizations tend to have team-specific technology procurement. This creates silos of tools and platforms, increases the cost, and lowers the effectiveness of knowledge sharing. On the contrary, centralizing procurement will be in line with the organization's tech landscape, align with future roadmap, provide better negotiation power with vendors, help in standardization of usage, and make it easier to enforce information security and other compliance guidelines.

The decision on which tools to procure should be based on thorough analysis and proper value articulation. Some aspects of tool selection and subsequently scaling the adoption are explained later in this whitepaper.

Another important aspect is subscriptions and license management. Uncontrolled allocation of subscriptions and licenses leads to cost leakage. Therefore, a mechanism for allocating and tracking the usage and benefits is necessary.

Challenges

Various stakeholders may worry about different aspects of AI intervention in their respective areas of responsibilities in the SDLC:

1. IT leaders and departmental heads worry about budget cuts in the following year due to optimization achieved in the current year.
2. Resistance may arise due to inter-departmental barriers.
3. Junior employees may fear job loss due to automation.

Remedies

Effectively managing various stakeholders at different levels requires policy changes and effective employee engagement:

1. The IT leaders and department heads should be incentivized to bring in AI-led or any technology-led efficiency gains. These are the kind of leaders who should be rewarded with more IT responsibilities to realize even more value.

2. Provide sufficient learning sessions to ensure that leaders are equipped to think holistically beyond just one tool being infused in one phase. They should consider multiple infusions at multiple phases and adjust operational, planning, and deployment aspects in tandem to ensure value realization.

3. There must be an overarching/central AI core team that breaks down the barriers between departments and enables easy adoption of AI in SDLC.

4. The fear factor in employees at junior levels can be mitigated by instilling a *"do more with less"* mindset. It will also help CxOs realign responsibilities with leaders who are able to achieve *'do more with less'* in every letter and spirit.

4. Recommendations on AI Adoption

Here's a stepwise guide to help your AI adoption journey. These recommendations are based on best-fit strategies that are proven among a diverse customer base.

a. Experiment and Find What Fits

Experiment

Understand the benefits for the organization

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Conduct experiments and POCs

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Requirements, setup, ISG, and usage guidelines

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Finalize from available options

Organizations must actively explore and validate the potential for applying AI within their SDLC. This is crucial for uncovering real-world scenarios where AI tools can work for an organization.

After identifying potential applications, such as AI Pair Programming or AI-Assisted Design Document creation, experiments should be performed and proofs of concepts developed. This will help in identifying the tools, LLMs, or approaches that best align with an enterprise landscape.

While these experiments are conducted, they must be vetted by the IT security team and, if necessary, the legal team. Usage guidelines and necessary (network, firewall) setup should be established as the organization moves beyond experimentation and POCs.

A tool that leverages an already approved LLM or hyperscaler stack can speed up the vetting process.

b. Onboard and Pilot

Once the tool or option is identified, an efficient license management mechanism needs to be established to ramp up and track usage. This will help with compliance and cost optimization.

The next step is to start onboarding users. Conducting training sessions, hackathons, and hands-on workshops can help empower users. These should be targeted based on user role. For example, AI Pair Programming sessions for developers and automation testers.

As users get familiar, it is important to assess the productivity and quality gains. For this, KPIs and baseline needs should be established, including defining measurement mechanism(s). These will form the framework to evaluate the effectiveness of Gen AI tools and eventually the ROI.

Finally, specific projects or teams should be identified for the pilot. They will provide valuable insights into practical applications and challenges with the AI/Gen AI tools. These insights will not only help validate and refine the expected ROI; it will also provide valuable inputs for fine-tuning the implementation strategy before a full-scale rollout.

Onboard

User onboarding, license management process

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User enablement - training, hackathons, and hands-on sessions

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Establish KPIs, ROI, baseline, and measurement mechanism

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Pilot with identified projects/teams

c. Scale and Soar

Scale

Plan and execute org-level rollout

Identify and train champions to drive penetration

Learn and implement best practices

Track usage increase, publish success stories

The insights gained from the pilot should be carefully considered to create a plan for a comprehensive rollout across the organization. While executing the plan, coordination across all departments and clear communication to all stakeholders are crucial.

Champions must be Identified across departments to act as evangelists. These champions should be trained to promote the tool and support effective usage within their departments.

Feedback should be gathered, and the performance of the tools must be monitored continuously. Best practices and success stories should be highlighted across the organization, leading to a culture of continuous learning and motivating further adoption.

Another key aspect is to regularly track and publish usage and impact details, highlighting positive outcomes and ROI.

Benefits and ROI for development will be different from testing and other phases. Hence, it is important to track these metrics for different use cases across the SDLC phases.

d. Sustain and Continuously Improve

To ensure the ongoing success of such adoption, maintaining high usage levels is paramount. Regularly monitoring statistics, aiming for at least 80% active utilization is important. Engagement strategies like check-ins, incentives, and training refreshers to encourage consistent usage are useful.

Obstacles that hinder usage need to be identified and addressed. This could range from a technical problem to reluctance to adapt, or even minimal success in a specific SDLC phase.

A strong feedback loop for reporting challenges and an efficient support mechanism will improve sustainability over time.

Finally, as Gen AI is a fast-evolving area, newer advancements, capabilities and features should be regularly assessed for potential use.

Sustain

Maintain over 80% usage across the enterprise

Identify and address any roadblocks to continuous utilization

Analyze team feedback for improvements and deficiencies

Evaluate new features and implement as appropriate

5. Benefits and Key Influencing Factors

The benefits realized in each phase are not standard across the board and are highly contextualized. The illustration below depicts the potential productivity benefits and the factors influencing them.

Potential Productivity Improvement in each SDLC Phase

Requirement Analysis	Design	Build	Test
Up to **20%**	Up to **15%**	Up to **30%**	Up to **30%**

Context-specific factors Influencing the benefits and ROI

SDLC Phase
Requirement Analysis, Design, Coding, Testing

Gen AI Tools
GitHub Copilot, AWS Q, Google Gemini, etc.

Project Type
Greenfield, Brownfield, Transformation, Migration, Steady-State Maintenance, etc.

Programming Language
Java, .NET, Python, etc.

Choice of LLMs
Size, open-source vs proprietary

Ongoing Cost
Subscription cost (for hosted LLMs), infrastructure cost (for on-prem LLMs)

Quality of Inputs
Clarity of instructions, the richness of context (documents, code, etc.), user experience, and awareness

The productivity benefits, along with other organizational ecosystem changes outlined in this whitepaper have great potential to generate value and ROI.

Conclusion

Transitioning from traditional SDLC to new-age ADLC is not just about adopting AI tools but rethinking processes, technology, and stakeholder management. In SDLC, the benefits presented by AI tools are tangible and immediate. However, realizing these benefits requires proper adoption through a comprehensive approach, making necessary adjustments in the ecosystem, and investing appropriately. The blueprint for ADLC adoption given in this white paper provides comprehensive insights to those looking to leverage AI to move from SDLC to ADLC.

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